

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a diverse range of synthesis and polymerization methods. The extracted data from multiple patents demonstrates various approaches to producing NBR with low acrylonitrile content. Key parameters such as conversion rates, temperatures, and emulsifier concentrations are highlighted across different examples. Recurring variables include acrylonitrile content, styrene concentration, and reaction temperatures. Trends indicate a focus on optimizing reaction conditions to achieve high conversion rates and desirable material properties. The patents also showcase different jurisdictions, including the US and EP, with publication years ranging from 2008 to 2017.

2. Research Scope & Selection Method

The patents included in this report were selected based on their direct relevance to the synthesis and polymerization of Low Acrylonitrile NBR. Only patents that demonstrated explicit methods for producing NBR with low acrylonitrile content were considered. The selection process involved a thorough review of patent titles, abstracts, and example descriptions to ensure that the included patents met the specified criteria.

3. Patent-by-Patent Analysis

PATENT EP2463335B1

Example GENERAL PROCEDURE 1

Parameter	Value	U	Context
Per	30		
Sol	70		
Viscosity	100		

Conversion	98		
Methyl	80		
Temperature	20		
Acrylonitrile	45		
2-vinylpyridine	2.2		
Sodium	2		
Emulsifier	20		

PATENT US10266615B2

Example EXAMPLE

Parameter	V	U	Context
Conversion	95		
Total	100		
Maturation	1		
##### Example EXAMPLE 3			
Parameter	Value	Unit	Context
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When	82		
##### Example EXAMPLE 4			

Parameter	Value	Unit	Context
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When	82		

PATENT EP3239198B1

Example EXAMPLE

Parameter	Value	Unit	Context
Ion	20		
Potassium	1.5		
Until	75		
Conversion	45		
Emulsifier	45		
Acrylonitrile	10		
Styrene	74		
Up	1		
Temperature	20		
During	45		
Below	20		

PATENT US10208151B2

Example EXAMPLE

Parameter	Value	U	Context
Conversion	97		
Up	1		
Temperature	1		

PATENT US8299175B2

Example EXAMPLE 1

Parameter	Value	U	Context
Ion	10		
Potassium	0.1		
Sodium	0.1		
Tert-dodecyl	0.1		
Emulsifier	0.1		
Conversion	97.5		
Toluene	100		
Temperature	48		
Ferrous	0.005		
Formaldehyde	0.23		

Styrene	9.6		
Acrylonitrile	3.0		
T-dodecyl	0.1		
Diisopropylenehydroperoxide	0.1		
Time	97.5		
Solid Content	0.2		

4. Cross-Patent Comparison

Comparing the ranges of key parameters across the different patents: - Acrylonitrile content: 3.0% (US8299175B2) to 45% (EP2463335B1) - Conversion rates: 45% (EP3239198B1) to 98% (EP2463335B1) - Reaction temperatures: 1°C (US10208151B2) to 48°C (US8299175B2) - Emulsifier concentrations: 0.1% (US8299175B2) to 45% (EP3239198B1)

5. Key Technical Trends

Technical trends across the analyzed patents include: - Optimization of reaction conditions to achieve high conversion rates and desirable material properties - Variation in acrylonitrile content to produce Low Acrylonitrile NBR - Use of different emulsifiers and concentrations to control reaction kinetics - Employment of various temperatures and reaction times to influence polymer properties

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year
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EP2463335B1	Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked rubber material, and method for producing nitrile rubber composition	Not disclosed	EP	2010-08-04
US10266615B2	Method for preparing acrylonitrile-butadiene-styrene graft copolymer and acrylonitrile-butadiene-styrene thermoplastic resin comprising the same	Not disclosed	US	2017-05-26
US8299175B2	Method of preparing thermoplastic resin having superior gloss, impact strength and whiteness	Not disclosed	US	2008-11-05

7. Excluded / Rejected Patents

Patent Number	Exclusion Reason
None	

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
EP2463335B1	Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked rubber material, and method for producing nitrile rubber composition	Not disclosed	EP	2010-08-04	https://patents.google.com/patent/EP2463335B1
US10266615B2	Method for preparing acrylonitrile-butadiene-styrene graft copolymer and acrylonitrile-butadiene-styrene thermoplastic resin comprising the same	Not disclosed	US	2017-05-26	https://patents.google.com/patent/US10266615B2

EP3239198B1	Method for preparing diene-based rubber latex and acrylonitrile-butadiene graft copolymer comprising the same	Not disclosed	EP	2015-12-24	https://patents.google.com
US10208151B2	Method for preparing large-size diene-based rubber latex and acrylonitrile-butadiene-styrene graft copolymer comprising the same	Not disclosed	US	2015-12-10	https://patents.google.com
US8299175B2	Method of preparing thermoplastic resin having superior gloss, impact strength and whiteness	Not disclosed	US	2008-11-05	https://patents.google.com